



Python Assignment 2

Goal:

To understand how different Python operators work

Part A: Relational Operators

(<, >, <=, >=, ==, !=)

1. Take two numbers as input and print:

```
num1 = int(input("Enter first number: "))
num2 = int(input("Enter second number: "))
print(num1 < num2)
print(num1 > num2)
print(num1 == num2)
print(num1 != num2)
print(num1 <= num2)
print(num1 >= num2)
```

👉 Try entering different values and write what each result means.

2. Input your **age** and check:

```
age = int(input("Enter your age: "))
print(age > 18)
print(age == 18)
print(age < 18)
```

3. Input your **marks** and test:

```
marks = int(input("Enter your marks: "))
print(20 < marks < 40)
```

👉 Try entering marks as 10, 30, and 50 – observe the results.

Part B: Logical Operators

(and, or, not)

4. Input two numbers and test:

```
a = int(input("Enter first number: "))
b = int(input("Enter second number: "))
print(a > 0 and b > 0)
print(a > 0 or b > 0)
print(not (a > b))
```

5. Input marks and full marks (FM):

```
marks = int(input("Enter marks: "))
FM = int(input("Enter full marks: "))
print(marks < 0 or marks > FM)
```

👉 Try values like 50 (out of 100) and 110 (out of 100).

6. Practice expressions:

```
print(5 > 3 and 2 < 4)
print(5 < 2 or 10 > 3)
print(not 7 == 7)
```

Part C: String Operators

(+ for joining, * for repeating)

7. Input your first and last name and print:

```
first_name = input("Enter first name: ")
last_name = input("Enter last name: ")
```

```
print(first_name + last_name)
print(first_name + " " + last_name)
```

8. Input a name and a number, and repeat the name:

```
name = input("Enter your name: ")
n = int(input("Enter how many times to repeat: "))
print(name * n)
```

Part D: Length Operator

(len)

9. Input your name and print its length:

```
name = input("Enter your name: ")
print(len(name))
```

10. Input any word and check if it's longer than 5 characters:

```
word = input("Enter a word: ")
print(len(word) > 5)
```

Part E: Membership Operator

(in)

11. Input a short sentence and a word to search:

```
text = input("Enter a sentence: ")
word = input("Enter a word to search: ")
print(word in text)
```

12. Try these and observe:

```
print("a" in "apple")
print("x" in "apple")
print("ram" in "ramhari")
print("ram " in "ram hari")
```

Part F: Identity Operator

(is, is not)

13. Try this:

```
a = 5
b = 5
print(a is b)
print(a is not b)
```

14. Try this example:

```
a = None
b = None
print(a is b)
print(a is not b)
```

15. Test identity on strings:

```
name1 = "Ram"
name2 = "Ram"
print(name1 is name2)
print(name1 == name2)
```

Part G: Variable Naming Practice

16. Write **5 valid** variable names and **5 invalid** ones.

Example:

```
✓ student_name = "Hari"
✗ 1student = "Hari"
```

17. Rename the following variables to meaningful names:

```
a = 20
b = 30
c = a + b
print(c)
```

👉 Replace with clear names like `math_marks`, `science_marks`, `total_marks`.

18. Try writing variable names in **snake_case**:

- `student_name`
 - `total_marks`
 - `number_of_students`
 - `full_name`
 - `english_marks`
-

19. Try assigning reserved words as variable names (e.g., `or = 5`) and observe the syntax error.

20. Observe these results and write down what happens:

```
print(10 > 5 and 5 < 10)
print(10 == 5 or 5 < 10)
print(not 3 > 2)
print("a" in "data")
print(len("python") >= 6)
```